to further it.’ Reflecting this concern, the three major trusts of Tata Trusts spent nearly 1,500 crore on fighting cancer in the years (2017–18 to 2021–22) alone.

Making Healthcare Affordable for All

One of the most important initiatives that the MIT Tata Center has undertaken and one which can have an enormous social impact is the development of low-cost medical devices to make healthcare affordable to the poor. Producing them was a challenge. It required the harnessing of the intellectual resources of the US institute to provide alternatives to the highly priced equipment manufactured by a few multinationals.

‘We brought in people from various domains, and we started questioning how we could get a blood test without penetrating the skin? How could we fix dental cavities in a different form, how do we have a low-cost eye testing machine that we could send to the villages and provide glasses and how do we get early warning on some of the diseases,’ were the questions for which they were seeking answers, Ratan says. This had inspiring results.

Skin imaging could be achieved with the help of a smartphone. It is highly mobile and can be taken to places where there is no electricity, or where its supply is erratic, and can identify cancerous lesions. This holds immense promise for the rural populace who live far away from hospitals and find it difficult to visit such medical facilities.

One of Ratan’s focus areas has been the manufacture of low-cost MRI diagnostic machines to fill a critical gap in the Indian health infrastructure. The more popular and powerful Tesla (the unit of measurement of the power of magnet in a machine) machines are generally manufactured by two large multinationals, GE and the German company Siemens. As they are oligopolistic products, they tend to be highly priced.

It is normal for the manufacturers to buy back the lower models of the machines (such as 1.5 Tesla MRIs) and replace them with the more expensive and powerful models (3 Tesla, for instance), placing them beyond the reach of most medical centres outside large cities and towns where the vast majority of Indians live. Ratan stepped in to find a way to make them more accessible to the larger section of the population. He opines: ‘And what we are saying is that many of the smaller hospitals can’t afford 3 Tesla Machines … And those older units could easily serve 70–80 per cent of the